

# PURBANCHAL UNIVERSITY

2026

Master of Computer Application (M.C.A.)/First Semester/ Final

Time: 03:00 hrs.

Full Marks: 60 / Pass Marks: 24

**MCA104/MCA114: Advanced Database Management System (New/Old)**

*Candidates are required to give their answers in their own words as far as practicable. The figures in the margin indicate full marks.*

## Group A

**Answer TWO questions.**

**2×12=24**

- 1(a) What is Lock Based Protocol? How is it performed? Write down its pitfalls.
- (b) Explain Indexed Based Algorithms with its benefits.
2. What is Serializable schedule? Explain serializability and its types. Explain testing of serializability with example.
3. Consider following schemas: 12  
Doctors (Did, Dname, Department, Address, Salary)  
Patients (Pid, Pname, Address, Age, Gender)  
Hospitals (Did, Pid, Hname, Location)  
Write SQL statements for
  - (i) Display ID of patient admitted to hospital at Patan and whose name starts with 'A'.
  - (ii) Delete the records of doctors whose salary is greater than average salary of doctors.
  - (iii) Increase salary of doctors by 20% who works in Emergency department.
  - (iv) Find average salary of doctors for those department whose average salary is greater then 80K.
  - (v) Find name of doctors as well as their hospitals name.
  - (vi) Find Maximum salary and Minimum salary of doctors department wise.

## Group B

**Answer SIX questions.**

**6×6=36**

4. Diagrammatically illustrate and discuss the steps involved in processing a query.
5. Explain JDBC and XML along with their benefits. Also provide suitable examples.
6. What do you mean by database audit trail? Briefly discuss about the best security practices on database.

**Contd. ...**



(2)

7. What is RAID (Redundant Array of Inexpensive Disks)? Explain its levels. -
8. Explain Log based recovery scheme along with deferred modification and immediate modification policies. -
9. Outline the concept of multimedia database, mobile database, semantic database and open-source database. \*
10. Consider the relational database:

employee (person-name, Street, city)

works (person-name, bank-name, salary)

bank (bank-name, city)

manages(person-name, manager-name)

Write relational algebraic expression for following tasks:

- (i) Find the names of all employees in database who live in same city as the company for which they work.
  - (ii) List average salary of employee company wise.
  - (iii) Modify the database so that Ram now lives in Kathmandu.
  - (iv) Delete all tuples in works relation for employee of XYZ Bank Ltd.
11. Suppose you are given the following requirements for a simple database for National Cricket League(NCL):
- the NCL has many teams
  - each team has a name, a city, a coach, a captain and set of players
  - each player belong to only one team
  - each player has a name, type(like batsman or bowler), a skill level, and a set of records
  - a team captain is also a player
  - each player is sponsored by at least one brand
  - a brand has its name, established date, property, multiple contact numbers. construct a clean and concise ER diagram indicating cardinality mappings as well as any role indicators.

~